

# upgrade.md — ब्रम्ह upgrade log (what the crawler improved in its own engine)

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Written when the ब्रम्ह crawler/scrapper or the self engine upgrades its own capability. Records *what is improved* in the machine-level core.

## Upgrade entries (newest first)

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- 1784183231: added `assembly/sakum_bramann.s` — from-scratch raw HTTP crawler
- HTML scraper + binary-hash quantum-learn loop (ब्रम्ह activity).
- 1784183231: added `assembly/sakum_webhook.s` — from-scratch raw assembly webhook receiver (POST /update) with a nerve-bus signal + bot-cycle dispatch.

## Pending upgrades

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- Multi-sphere parallelism (crawl several `Knowledge/` subtrees per pulse).
- Quantum-weighted sphere selection (pick next sphere by learned hash entropy).
- Scraper: extract `<meta>`, JSON-LD, and math (LaTeX) nodes for the KB.
- 1784185580: added (auto-generated library for vector)
- 1784185692: added `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_bounds.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_bounds.sakum` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_numeric.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_numeric.sakum` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_overflow.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_overflow.sakum` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_vector.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_vector.sakum` (auto-generated library for vector)
- 1784185831: added `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_bounds.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_bounds.sakum` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_numeric.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_numeric.sakum` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_overflow.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_overflow.sakum` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_vector.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_vector.sakum` (auto-generated library for vector)

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## Website build findings — `site/` (learned 1784205000)

Built a complete website (Tailwind + KaTeX + an in-browser Sakum interpreter) and a real Sakum app ( `site/app.sakum` , a heart-rate survivability simulator using `नाम/क्रिया/यदि/यावत्/वेक्टर/latex/पल्स` ). Findings below are concrete **language-model improvements** for `SAKUM_LANG.md` and the `self` engine.

### F1 — Keyword spelling is inconsistent across the corpus (HIGH)

- `hello.sakum` uses `प्रत्यागम` for return; `examples/lib_bounds.sakum` uses `वापस` . `lib_bounds.sakum` uses `चर` for `let` while the spec + `hello.sakum` use `नाम` . `मुद्रण` vs `लेख` for print.
- **Fix:** `SAKUM_LANG.md` §5 must declare ONE canonical Devanagari keyword per concept and forbid aliases in source. The interpreter should accept aliases only inside `assembly/` machine comments, never as parseable tokens. Add a `self` lint pass that rejects unknown/duplicate keywords (recorded as a `mistake` in `memory.md`).

### F2 — `for` loop is specified but no example exercises it (MED)

- `पर्यन्तम्` / `for` exists in the keyword table but no `.sakum` file uses it; the interpreter implemented it blind. The website playground has no `for` sample.
- **Fix:** add `examples/loop.sakum` covering `for` , `while` , nested loops, and `range()`. The `self` engine should generate a `for` smoke-test each pulse.

### F3 — Builtins are documented but not all are implemented anywhere (MED)

- `SAKUM_LANG.md` lists `sin/cos/tan/deriv/integrate/gcd/fib/gamma/zeta/...` and `simd(n) / simd_info()` , but `assembly/sakum_eval.s` only does `let/expr` . The math richness lives only in prose ( `math100.sakum` ) with no executing backend.
- **Fix:** promote the website interpreter's builtin set to the spec's source of truth, OR clearly mark which builtins are "spec intent, not yet in a backend". Add a `self` coverage table: builtin -> implemented-in (eval | wasm | site).

### F4 — No standard library module system; everything is top-level (HIGH)

- Real programs need `import` . Building a multi-feature site app forced all logic into one file. `lib_*.sakum` files exist but are standalone, not importable.
- **Fix:** add `आयात / import "path"` to `SAKUM_LANG.md` §1.1 and the front-end. The `self` engine should resolve `lib_*.sakum` via a module cache.

## F5 — Vectors lack an explicit element type and indexing is undocumented (MED)

- वेक्टर / `vektor` is specified, but element indexing `D[i]` and scalar broadcast `v * 2` only appear implicitly. The interpreter had to infer broadcast/dot/cross semantics.
- **Fix:** document `D[i]` read/write, `dot/cross/norm`, and broadcast rules explicitly in §1.11. Add `len(v)` and `v[i] = x` to the spec.

## F6 — The `self` engine has no typed value model across backends (MED)

- `sakum_eval.s` stores vars as 32-bit ints; the website interpreter uses JS dynamic values; `sakum_wasm.s` uses WASM i32/f32. The same source means different things per backend.
- **Fix:** SAKUM\_LANG.md §1.1 should define a canonical value lattice (i64, f64, bool, vec, str, fn) that every backend must lower to identically. Add a cross-backend conformance test in `self/patches/`.

## F7 — Error messages are not user-facing (LOW)

- The interpreter throws `parse: expected ' ) '` style messages; spec wants व्याख्या (vyakhya) to explain error codes in human language.
- **Fix:** wrap parse/runtime errors and route them through a व्याख्या explainer that emits both the code and a Devanagari + English hint.

## F8 — Survivability score needs a defined formula (LOW)

- `memory.md` stores `survive:` and `mistake` lines but the website had to guess a percentage. There is no documented survivability formula.
- **Fix:** define `survivability = survive / (survive + mistakes)` in §1.9 and have `self` write it explicitly each cycle so dashboards read one value.

## Action taken by this build

- Added `site/` (`index.html` + `app/sakum.js` + `app/site.js`) — a working reference app + interpreter proving the front-end subset (`let/fn/if/else/while/for`, vectors, builtins, latex, pulse) is implementable and runnable.
- Recorded F1–F8 as pending language-model improvements above.
- 1784202315: added `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_bounds.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_bounds.sakum` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_numeric.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_numeric.sakum` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/assembly/sakum_lib_overflow.s` `/Users/Amit/Downloads/ Vendor Quotations/untitled folder 2/Sakum Lang/examples/lib_overflow.sakum` `/Users/Amit/`



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- 1784234807: added (auto-generated library for vector)
- 1784289700: added examples/lib\_crud.sakum (Sakum CRUD store library: push/store\_new/store\_add/store\_get/store\_update/store\_delete/store\_count/store\_max\_id/record\_total/store\_value) + examples/dashboard.sakum + examples/dashboard\_full.sakum (self-contained CRUD dashboard with self-engine pulse metrics). Grew the in-language library with real, runnable CRUD functions; exercised learn+update cycle.